

A Comparison of LSTM and BiLSTM for Forecasting the Air Pollution Index and Meteorological Conditions in Jakarta

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Abstract—Jakarta is the capital city of Indonesia where air pollution becomes one of the problems that must be properly handled. The historical data of the air pollution index is beneficial for developing models for forecasting future values. One of the advantages of forecasting air pollution is to help people to arrange future plans to reduce the dangerous effect on health. Analyzing a record of meteorological conditions can be used to understand climate change. This paper reports the comparison of Long Short Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) models for multivariate forecasting of the air pollution index and meteorological conditions in Jakarta. It also informs the performance of those algorithms for forecasting the observed variables before and during the Coronavirus disease (Covid-19) outbreak to analyze the effect of the pandemic on the environment. The experiments use a historical time series dataset from 2010 - 2021. The experimental results show that LSTM and BiLSTM work well to forecast PM₁₀, temperature, humidity, and wind speed. In this case study, there are no significant differences in the performance of LSTM and BiLSTM.

Index Terms—LSTM, BiLSTM, forecasting, air pollution, meteorology

I. INTRODUCTION

Air pollution is one of the common problems in society. Air pollution has a serious impact on human health [1] [2]. The ministry of environment and forestry in Indonesia measures air quality ambient using the Air Quality Index (AQI). The AQI is measured for 7 pollutants, i.e, PM₁₀(Particulate Matter 10), PM_{2.5} (Particulate Matter 2.5), SO₂ (Sulfur dioxide), CO (carbon monoxide), O₃ (Ozone), NO₂ (nitrogen dioxide), and HC (hydrocarbons). The Air Quality Index is categorized into Good (1-50), Moderate (51-100), Unhealthy (101-200), Very Unhealthy (201-300), and Hazardous (≥ 301) [3]. Air Quality Index relates to the health impact on public health [4]. The AQI can be used as a risk communication tool for respiratory health risks [5].

Climate change has several impacts on public health [6] [7]. Climate change is related to meteorological conditions. Recent

studies reveal the impact of the meteorological condition on the Covid-19 pandemic [8] [9] [10].

LSTM and BiLSTM have been successfully implemented for forecasting the time series datasets [11] [12]. LSTM is used for forecasting AQI and has been studied in previous research [13] [14] [15]. LSTM is also applied for forecasting meteorological conditions [16] [17] [18] [19]. BiLSTM has been implemented for traffic forecasting [20] and the prediction of the time series financial market [21].

Developing models for forecasting the air pollution index is beneficial for preparing plans to reduce the effect on human health and the environment. Meanwhile, the model for forecasting meteorological conditions is useful to forecast the weather. Weather forecasting has some benefits, i.e., protecting human lives, safety, and farming. The goal of this paper is to evaluate the performance of LSTM and BiLSTM in forecasting the air pollution index and meteorological conditions in Jakarta. This paper also reports the performance of those algorithms for forecasting the observed variables before and during the Covid-19 outbreak.

II. METHOD

LSTM network is a specific version of a recurrent neural network (RNN) [22] [23]. The LSTM network for forecasting receives the known data and outputs forecasting results O_t . The memory unit of LSTM has three gates: input gate, forget gate, and output gate. Let σ , S_t , X_t , S_{t-1} , $\tilde{S}_t$, S_t , and O_t be the activation function, the state of the cell, input data of every gate, the previous state of the memory cell, the new state of the memory cell, the final state of the memory cell, and the final output of the memory unit, respectively. The mathematical models for LSTM can be defined using equation (1 - 6), where $\circ$ is the Hadamard product, and i_t , f_t are the output of different gate [24]. $W^{(i)}$, $W^{(f)}$, $W^{(o)}$, $U^{(i)}$, $U^{(f)}$, $U^{(o)}$, and $U^{(c)}$ are coefficient matrices.

$$i_t = \sigma(W^{(i)}X_t + U^{(i)}S_{t-1}) \quad (1)$$

III. EXPERIMENTAL RESULTS

$$f_t = \sigma(W^{(f)}X_t + U^{(f)}S_{t-1}) \quad (2)$$

$$o_t = \sigma(W^{(o)}X_t + U^{(o)}S_{t-1}) \quad (3)$$

$$\tilde{S}_t = \tanh(W^{(c)}X_t + U^{(c)}S_{t-1}) \quad (4)$$

$$S_t = f_t \circ S_{t-1} + i_t \circ \tilde{S}_t \quad (5)$$

$$O_t = o_t \circ \tanh(S_t) \quad (6)$$

BiLSTM is an extended version of LSTM when two LSTM are applied to the input data [25]. The hidden layer output of BiLSTM contains the activation output of the forward and backward hidden layers. Suppose H_t , $\vec{h}_t$, $\overleftarrow{h}_t$ denote the hidden layer input, the output of forward structure, and the output of backward structure, respectively. The mathematical models of BiLSTM can be defined by equations (7), (8), and (9) [21]. In this paper, LSTM and BiLSTM are implemented for forecasting the air pollution index and meteorological conditions.

$$\vec{h}_t = \sigma(W_{x\vec{h}}x_t + W_{\vec{h}\vec{h}}\vec{h}_{t-1} + b_{\vec{h}}) \quad (7)$$

$$\overleftarrow{h}_t = \sigma(W_{x\overleftarrow{h}}x_t + W_{\overleftarrow{h}\overleftarrow{h}}\overleftarrow{h}_{t-1} + b_{\overleftarrow{h}}) \quad (8)$$

$$H_t = W_{x\vec{h}}\vec{h}_t + W_{x\overleftarrow{h}}\overleftarrow{h}_t + b_y \quad (9)$$

The forecasting results can be evaluated using mean absolute error (MAE), mean square error (MSE), root mean square error (RMSE), and coefficient of variation (CV). Suppose y' , y , σ , μ , and n represent the predicted value, the true value, the standard deviation of the dataset, the mean of the dataset, and the number of samples. MAE, MSE, RMSE, and CV can be computed using equations (10), (11), (12), and (13), respectively [26] [27].

$$MAE_{y',y} = \frac{1}{n} \sum_{i=1}^n |y'_i - y_i| \quad (10)$$

$$MSE_{y',y} = \frac{1}{n} \sum_{i=1}^n (y'_i - y_i)^2 \quad (11)$$

$$RMSE_{y',y} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y'_i - y_i)^2} \quad (12)$$

$$CV = \frac{\sigma}{\mu} \quad (13)$$

A. Data

The datasets used in this paper are the air pollution index and the meteorological conditions in Jakarta. The air pollution index dataset is downloaded from Open Data Jakarta [28]. The air pollution index dataset is a time series data from 2010 - 2021 containing 4383 samples and 5 variables (PM₁₀, SO₂, CO, O₃, and NO₂). The meteorological conditions dataset is obtained from online datasets belonging to BMKG [29]. The meteorological dataset is a daily record of average temperature (°C), humidity (%), rainfall (mm), sunshine duration (hours), and average wind speed (m/s) from 2010 - 2021. The meteorological dataset has 4383 samples.

B. Discussion

This paper runs LSTM and BiLSTM for forecasting with six different scenarios:

- Experiment 1 using air pollution index dataset (training set from 2010 - 2015 and testing set from 2016 - 2017)
- Experiment 2 using air pollution index dataset (training set from 2010 - 2017 and testing set from 2018 - 2019)
- Experiment 3 using air pollution index dataset (training set from 2010 - 2019 and testing set from 2020 - 2021)
- Experiment 4 using Meteorological conditions dataset (training set from 2010 - 2015 and testing set from 2016 - 2017)
- Experiment 5 using Meteorological conditions dataset (training set from 2010 - 2017 and testing set from 2018 - 2019)
- Experiment 6 using Meteorological conditions dataset (training set from 2010 - 2019 and testing set from 2020 - 2021)

The reason for running experiments using different scenarios is to examine the performance of LSTM and BiLSTM to forecast air pollution index and meteorological conditions before and after the Covid-19 outbreak. The forecasting using testing datasets before the year 2020 indicates the situation before the Covid-19 outbreak. The forecasting using the testing dataset from 2020 - 2021 shows the situation during the Covid-19 outbreak.

The experiments implement LSTM and BiLSTM from Tensorflow Keras [30]. The models consist of stacked layers with 128 and 64 units, a dropout layer, and a dense layer. The training step of LSTM and BiLSTM are run for 50 epochs. They implement the Softmax activation function. The forecasting results of the air pollution index are illustrated in Fig. 1. In general, LSTM and BiLSTM work well to forecast PM₁₀, CO, O₃, and NO₂ from 2016 - 2017. However, it looks difficult to forecast SO₂ appropriately. LSTM and BiLSTM produce forecasting values lower than the actual values for SO₂ from 2016 - 2017 but those values have a similar trend to the actual values. The forecasting results of PM₁₀ and NO₂ from 2018 - 2019 have smaller errors than errors of CO and O₃. The forecasting for the air pollution index from 2020 - 2021 for SO₂, CO, and NO₂ is less accurate. In 2020 - 2021,

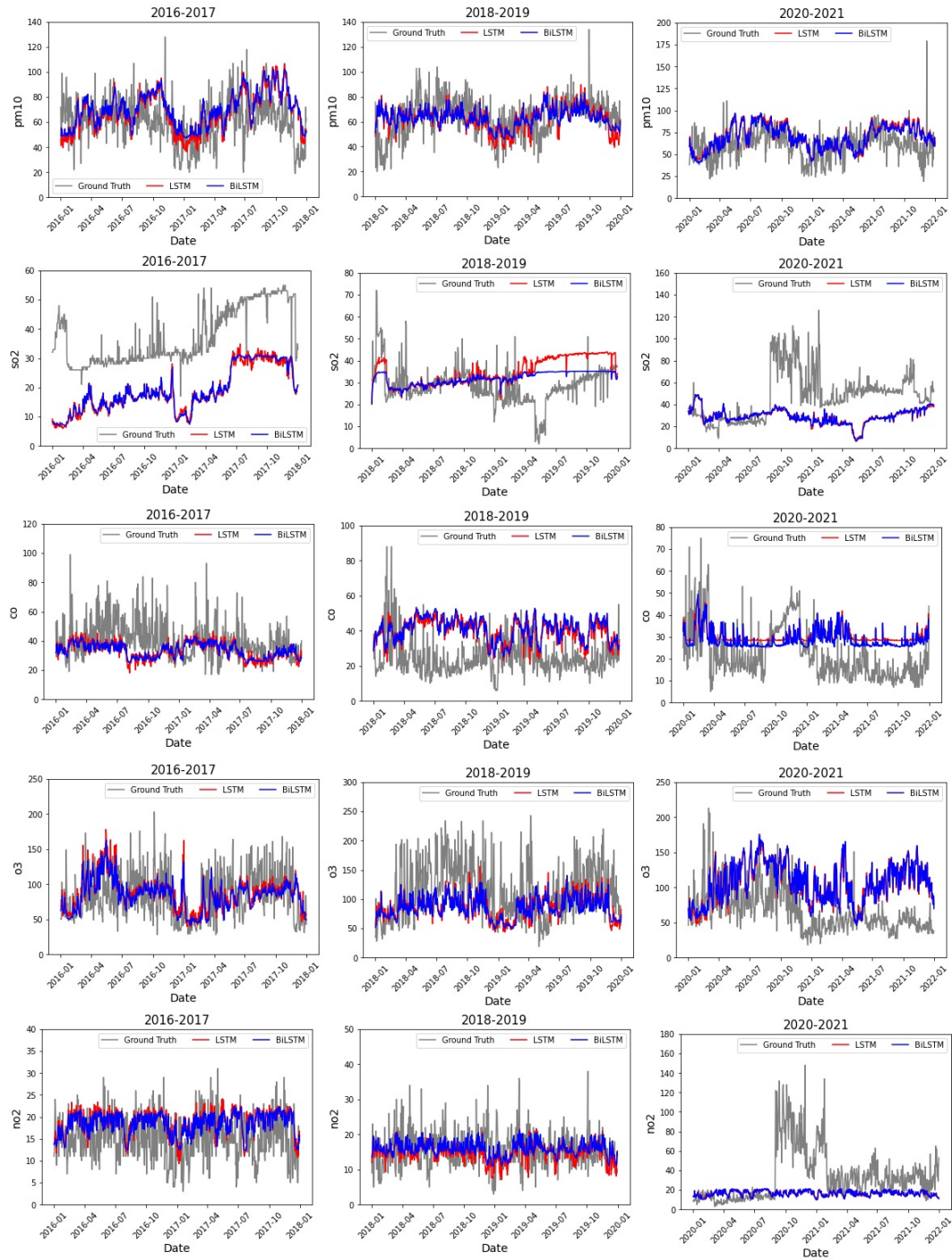


Fig. 1. A forecasting of air pollution index.

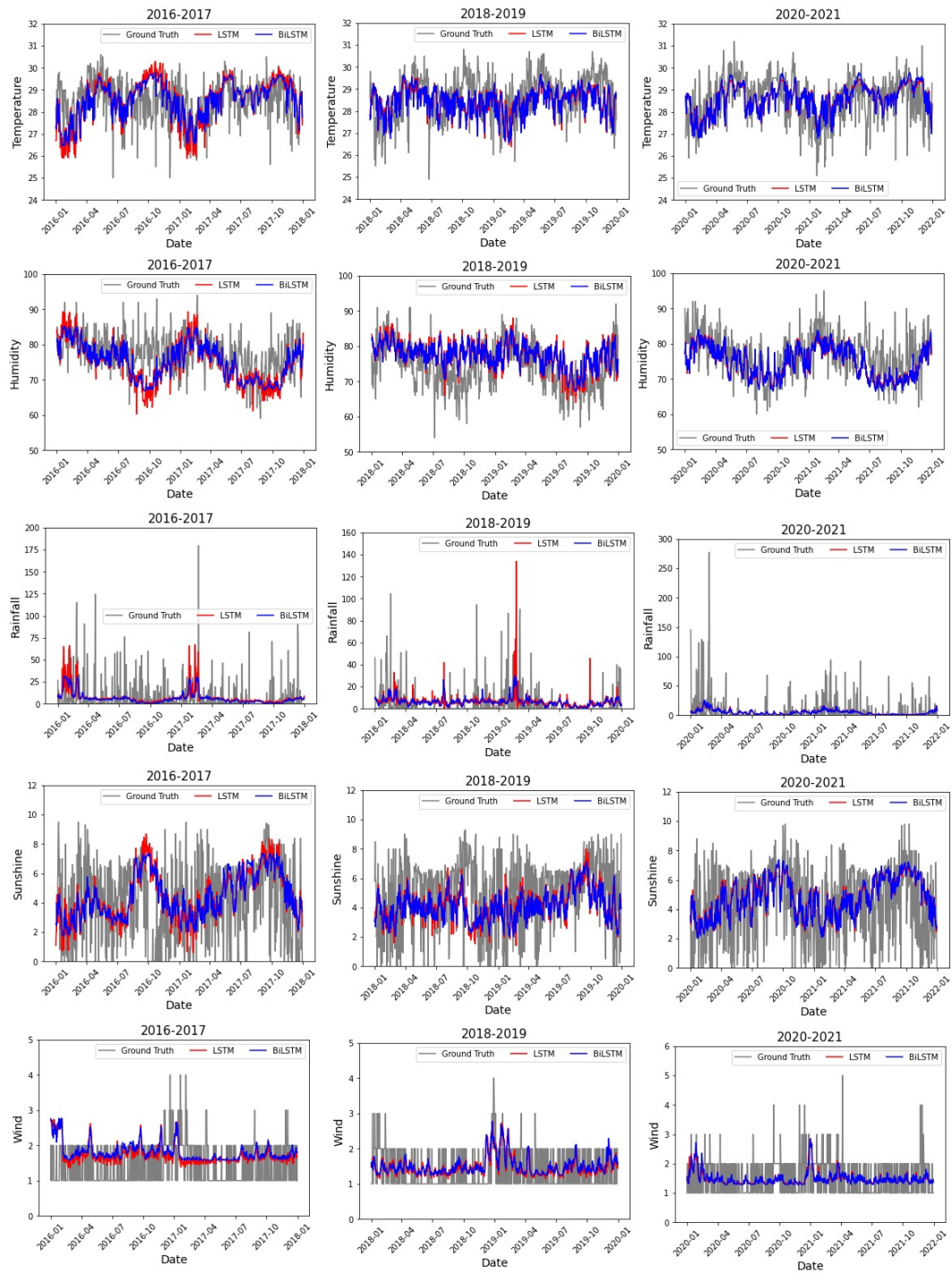


Fig. 2. A forecasting of meteorological conditions.

TABLE I
EVALUATION METRICS OF DATA FORECASTING FROM 2016 - 2017

Variables	MAE _L	MSE _L	RMSE _L	CV _L	MAE _B	MSE _B	RMSE _B	CV _B	Actual CV
PM ₁₀	16.64	423.61	20.58	0.22	17.12	443.78	21.07	0.20	0.28
SO ₂	18.76	394.12	19.85	0.40	18.58	385.00	19.62	0.40	0.26
CO	10.72	203.66	14.27	0.16	10.48	197.75	14.06	0.15	0.30
O ₃	27.12	1191.59	34.52	0.26	26.61	1153.60	33.96	0.25	0.38
NO ₂	5.15	40.12	6.33	0.14	4.96	37.53	6.13	0.14	0.32
Temperature	0.87	1.28	1.13	0.03	0.80	1.06	1.03	0.03	0.03
Humidity	5.23	44.49	6.67	0.08	4.89	38.66	6.62	0.07	0.07
Rainfall	9.31	294.36	17.16	1.24	8.45	257.01	16.03	1.05	2.41
Sunshine	2.25	8.08	2.84	0.36	2.24	7.82	2.80	0.35	0.64
Wind Speed	0.61	0.48	0.70	0.16	0.61	0.50	0.71	0.14	0.39

TABLE II
EVALUATION METRICS OF DATA FORECASTING FROM 2018 - 2019

Variables	MAE _L	MSE _L	RMSE _L	CV _L	MAE _B	MSE _B	RMSE _B	CV _B	Actual CV
PM ₁₀	12.74	267.73	16.36	0.15	12.07	243.80	15.61	0.11	0.24
SO ₂	9.16	140.90	11.87	0.17	6.79	89.99	9.49	0.09	0.29
CO	17.68	391.63	19.79	0.16	18.65	430.42	20.75	0.16	0.39
O ₃	40.58	2722.96	52.18	0.23	41.50	2844.31	53.33	0.21	0.40
NO ₂	4.27	30.21	5.50	0.17	4.09	27.73	5.27	0.13	0.31
Temperature	0.84	1.12	1.06	0.02	0.87	1.06	1.08	0.02	0.03
Humidity	5.30	45.80	6.77	0.05	5.26	38.66	6.67	0.05	0.08
Rainfall	7.58	162.87	12.76	1.03	7.14	257.01	11.81	0.62	2.79
Sunshine	2.08	6.39	2.53	0.25	2.10	7.82	2.54	0.26	0.48
Wind Speed	0.50	0.33	0.58	0.19	0.51	0.50	0.57	0.18	0.39

TABLE III
EVALUATION METRICS OF DATA FORECASTING FROM 2020 - 2021

Variables	MAE _L	MSE _L	RMSE _L	CV _L	MAE _B	MSE _B	RMSE _B	CV _B	Actual CV
PM ₁₀	16.18	400.55	20.01	0.20	15.43	374.22	19.34	0.20	0.25
SO ₂	22.23	855.17	29.24	0.23	22.71	849.64	29.15	0.24	0.44
CO	12.76	203.98	14.28	0.11	11.87	183.13	13.53	0.15	0.53
O ₃	47.26	3015.73	54.92	0.27	47.91	3095.87	55.64	0.27	0.44
NO ₂	20.73	984.18	31.37	0.16	20.75	985.57	31.39	0.17	0.79
Temperature	0.77	0.93	0.96	0.02	0.78	0.96	0.98	0.02	0.03
Humidity	4.55	33.49	5.79	0.06	4.59	34.38	5.86	0.06	0.08
Rainfall	8.32	377.60	19.43	0.83	8.14	377.39	19.43	0.92	2.82
Sunshine	2.03	6.42	2.53	0.28	2.04	6.36	2.52	0.28	0.57
Wind Speed	0.55	0.41	0.64	0.15	0.55	0.41	0.64	0.15	0.41

the level of CO decreased. The decrease in CO during the Covid-19 outbreak is associated with a decrease in economic growth and transport restrictions [31]. The previous studies found that the burning of fossil fuels contributes to increasing in NO₂ and SO₂ [32]. The less accurate prediction from 2020 - 2021 might cause by the Covid-19 outbreak which affected the activities in 2020. For instance, the explosion of SO₂ and NO₂ happened from September 2020 - January 2021. The exposure of SO₂ and NO₂ is dangerous for health [33] [34]. It needs further study to find the relationship between the Covid-19 outbreak and the air pollution index. Generally, the air pollutants in Jakarta have an increasing trend. The increasing trend of air pollution is one of the early warnings that the air pollution in Jakarta must be handled properly before it makes dangerous human life and environmental disaster.

The forecasting results of meteorological conditions are described in Fig. 2. LSTM and BiLSTM work well to forecast temperature, humidity, and wind speed, nonetheless, they are

less accurate to predict rainfall and the duration of sunshine. Jakarta has a tropical climate that typically has long sunshine hours. The duration of sunshine is usually affected by the presence of clouds. It seems other factors need to include for more accurate forecasting of rainfall and sunshine duration. This idea will be studied for future research. Before and during the Covid-19 outbreak, the trend of meteorological conditions in Jakarta remains the same.

The evaluation metrics are described in Table I, II, and III. The letter *L* and *B* refer to LSTM and BiLSTM, respectively. The forecasting of air pollution index before the Covid-19 outbreak produces slightly lower errors than that during the Covid-19 outbreak. The high errors are mostly produced by the prediction of SO₂, O₃, and NO₂. The performance of LSTM and BiLSTM to forecast meteorological data before and during the Covid-19 outbreak shows no significant difference.

IV. CONCLUSION

In conclusion, LSTM and BiLSTM work well to predict PM_{10} , temperature, humidity, and wind speed. This paper reports the comparison of LSTM and BiLSTM for forecasting the air pollution index and meteorological conditions based on time series data in Jakarta from 2010 - 2021. Generally, the forecasting results of the air pollution index before the Covid-19 outbreak produce smaller errors than that during the pandemic. The Covid-19 outbreak possibly contributes to the level of CO , SO_2 , and NO_2 where they have an unusual trend from 2020 - 2021. LSTM and BiLSTM are less accurate to forecast rainfall and sunshine duration.

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